ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 90°	Reason with the angle relations.
2	**(d)** 120°	Share 180° in the given ratio.
3	**(b)** 110°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
4	**(c)** 290°	A complete turn is 360°.
5	**(d)** 55°	Larger + smaller $= 90^\circ$ and larger − smaller $= d$.
6	**(a)** 80°	Larger + smaller $= 180^\circ$.
7	**(d)** 120°	Total $= 180^\circ$ in 3 equal parts.
8	**(b)** parallel	Reason with the angle relations.
9	**(a)** Both A and R are true and R is the correct explanation of A.	90 - 35 = 55.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but equality of vertically opposite angles does not explain the linear pair.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the angle be y: $180 - y = 3(90 - y)$; $2y = 90$ (1); $y = 45^\circ$ (1).	1+1
12	Let the complement be y; the angle is $3y + 6$ and $y + (3y + 6) = 90$; $4y = 84$, so $y = 21$ (1); The angle is 69° (1).	1+1
13	Larger $= (180 + 18) \div 2 = 99^\circ$; Smaller $= (180 - 18) \div 2 = 81^\circ$ (1); Complement of the smaller $= 90^\circ - 81^\circ = 9^\circ$ (1).	1+1
14	Let the angle be y: $180 - y = 4(90 - y)$; $3y = 180$ (1); $y = 60^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (35 + 45 + 50)^\circ = 50^\circ$ (1); Kinds: 35° acute, 45° acute, 50° acute, 50° acute (1).	1+1+1
16	One part $= 180^\circ \div 5 = 36^\circ$ (1); Smaller $= 36^\circ$ and larger $= 144^\circ$ (1); Complement of the smaller $= 54^\circ$ and supplement of the larger $= 36^\circ$ (1).	1+1+1
17	Other angle $= 180^\circ - 40^\circ = 140^\circ$ (1); Halves: 20° and 70° (1); Angle between the bisectors $= 20^\circ + 70^\circ = 90^\circ$ (1).	1+1+1
	OR: Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (20 + 75 + 30)^\circ = 55^\circ$ (1); Kinds: 20° acute, 75° acute, 30° acute, 55° acute (1).	1+1+1
18	One part $= 180^\circ \div 9 = 20^\circ$ (1); Smaller $= 20^\circ$ and larger $= 160^\circ$ (1); Complement of the smaller $= 70^\circ$ and supplement of the larger $= 20^\circ$ (1).	1+1+1
	OR: Other angle $= 180^\circ - 70^\circ = 110^\circ$ (1); Halves: 35° and 55° (1); Angle between the bisectors $= 35^\circ + 55^\circ = 90^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 65° (1). (ii) 65° (1). (iii) $180^\circ - 65^\circ = 115^\circ$ (1). (iv) Four angles are 65° (1) and four angles are 115° (1).	1+1+1+2
20	Total parts $= 9$ and $180^\circ \div 9 = 20^\circ$ (1); The angles are 40°, 60°, 80° (2); Kinds: 40° acute, 60° acute, 80° acute (2).	1+2+2
	OR: They are supplementary: one part $= 180^\circ \div 5 = 36^\circ$ (1); The angles are 36° and 144° (1); Four of the eight angles equal the smaller angle (vertically opposite, corresponding and alternate angles of it) (1); The other four equal 144° (linear pairs) (2).	1+1+1+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.

- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 225°	A complete turn is 360°.
2	**(b)** 60°	Larger + smaller $= 90^\circ$ and larger − smaller $= d$.
3	**(a)** 70°	Larger + smaller $= 180^\circ$.
4	**(a)** 135°	Total $= 180^\circ$ in 4 equal parts.
5	**(c)** 90°	Reason with the angle relations.
6	**(d)** 100°	Share 180° in the given ratio.
7	**(b)** 140°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
8	**(b)** 260°	A complete turn is 360°.
9	**(c)** A is true but R is false.	A is true. Vertically opposite angles are equal, they do not always add up to 180 degrees, so R is false.
10	**(d)** A is false but R is true.	Corresponding angles are equal, so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the complement be y; the angle is $3y + 10$ and $y + (3y + 10) = 90$; $4y = 80$, so $y = 20$ (1); The angle is 70° (1).	1+1
12	Larger $= (180 + 34) \div 2 = 107^\circ$; Smaller $= (180 - 34) \div 2 = 73^\circ$ (1); Complement of the smaller $= 90^\circ - 73^\circ = 17^\circ$ (1).	1+1
13	Let the angle be y: $180 - y = 6(90 - y)$; $5y = 360$ (1); $y = 72^\circ$ (1).	1+1
14	Let the complement be y; the angle is $3y + 14$ and $y + (3y + 14) = 90$; $4y = 76$, so $y = 19$ (1); The angle is 71° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (40 + 60 + 25)^\circ = 55^\circ$ (1); Kinds: 40° acute, 60° acute, 25° acute, 55° acute (1).	1+1+1
16	One part $= 180^\circ \div 10 = 18^\circ$ (1); Smaller $= 18^\circ$ and larger $= 162^\circ$ (1); Complement of the smaller $= 72^\circ$ and supplement of the larger $= 18^\circ$ (1).	1+1+1
17	Other angle $= 180^\circ - 100^\circ = 80^\circ$ (1); Halves: 50° and 40° (1); Angle between the bisectors $= 50^\circ + 40^\circ = 90^\circ$ (1).	1+1+1
	OR: Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (15 + 85 + 45)^\circ = 35^\circ$ (1); Kinds: 15° acute, 85° acute, 45° acute, 35° acute (1).	1+1+1
18	One part $= 180^\circ \div 15 = 12^\circ$ (1); Smaller $= 24^\circ$ and larger $= 156^\circ$ (1); Complement of the smaller $= 66^\circ$ and supplement of the larger $= 24^\circ$ (1).	1+1+1
	OR: Other angle $= 180^\circ - 110^\circ = 70^\circ$ (1); Halves: 55° and 35° (1); Angle between the bisectors $= 55^\circ + 35^\circ = 90^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 40° (1). (ii) $180^\circ - 40^\circ = 140^\circ$ (1). (iii) 360° (1). (iv) They form a linear pair, so $5x - 10 = 180$ and $x = 38$ (1); the angles are 86° and 94° (1).	1+1+1+2
20	Total parts $= 12$ and $180^\circ \div 12 = 15^\circ$ (1); The angles are 30°, 45°, 105° (2); Kinds: 30° acute, 45° acute, 105° obtuse (2).	1+2+2
	OR: They are supplementary: one part $= 180^\circ \div 12 = 15^\circ$ (1); The angles are 15° and 165° (1); Four of the eight angles equal the smaller angle (vertically opposite, corresponding and alternate angles of it) (1); The other four equal 165° (linear pairs) (2).	1+1+1+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.

- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 144°	Total $= 180^\circ$ in 5 equal parts.
2	**(a)** 45°	Reason with the angle relations.
3	**(c)** 150°	Share 180° in the given ratio.
4	**(b)** 155°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
5	**(c)** 335°	A complete turn is 360°.
6	**(c)** 65°	Larger + smaller $= 90^\circ$ and larger − smaller $= d$.
7	**(c)** 40°	Larger + smaller $= 180^\circ$.
8	**(a)** 150°	Total $= 180^\circ$ in 6 equal parts.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R is the test for parallel lines.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	110 + 70 = 180. Both true, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Larger $= (180 + 46) \div 2 = 113^\circ$; Smaller $= (180 - 46) \div 2 = 67^\circ$ (1); Complement of the smaller $= 90^\circ - 67^\circ = 23^\circ$ (1).	1+1
12	Let the angle be y: $180 - y = 7(90 - y)$; $6y = 450$ (1); $y = 75^\circ$ (1).	1+1
13	Let the complement be y; the angle is $3y + 18$ and $y + (3y + 18) = 90$; $4y = 72$, so $y = 18$ (1); The angle is 72° (1).	1+1
14	Larger $= (180 + 72) \div 2 = 126^\circ$; Smaller $= (180 - 72) \div 2 = 54^\circ$ (1); Complement of the smaller $= 90^\circ - 54^\circ = 36^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (30 + 30 + 70)^\circ = 50^\circ$ (1); Kinds: 30° acute, 30° acute, 70° acute, 50° acute (1).	1+1+1
16	One part $= 180^\circ \div 10 = 18^\circ$ (1); Smaller $= 54^\circ$ and larger $= 126^\circ$ (1); Complement of the smaller $= 36^\circ$ and supplement of the larger $= 54^\circ$ (1).	1+1+1
17	Other angle $= 180^\circ - 130^\circ = 50^\circ$ (1); Halves: 65° and 25° (1); Angle between the bisectors $= 65^\circ + 25^\circ = 90^\circ$ (1).	1+1+1
	OR: Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (55 + 25 + 65)^\circ = 35^\circ$ (1); Kinds: 55° acute, 25° acute, 65° acute, 35° acute (1).	1+1+1
18	One part $= 180^\circ \div 15 = 12^\circ$ (1); Smaller $= 12^\circ$ and larger $= 168^\circ$ (1); Complement of the smaller $= 78^\circ$ and supplement of the larger $= 12^\circ$ (1).	1+1+1
	OR: Other angle $= 180^\circ - 150^\circ = 30^\circ$ (1); Halves: 75° and 15° (1); Angle between the bisectors $= 75^\circ + 15^\circ = 90^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $90^\circ - 70^\circ = 20^\circ$ (1). (ii) $180^\circ - 70^\circ = 110^\circ$ (1). (iii) Yes, because $40^\circ + 50^\circ = 90^\circ$ (1). (iv) Let the angle be x: $x = (90 - x) + 20$ gives $x = 55^\circ$ (1); its complement is 35° (1).	1+1+1+2
20	Total parts $= 12$ and $180^\circ \div 12 = 15^\circ$ (1); The angles are 45°, 60°, 75° (2); Kinds: 45° acute, 60° acute, 75° acute (2).	1+2+2
	OR: They are supplementary: one part $= 180^\circ \div 18 = 10^\circ$ (1); The angles are 10° and 170° (1); Four of the eight angles equal the smaller angle (vertically opposite, corresponding and alternate angles of it) (1); The other four equal 170° (linear pairs) (2).	1+1+1+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.

- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

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